

Curriculum Vol 0 Chapter 1 — D_{IV}^5 Autogenic Proto-Geometry (Chapter-Grade Draft v0.4 — Cross-Cartan three-pillar + substrate self-amenability absorption)

Keeper (original) + Lyra (Friday v0.3→v0.4 prose depth-investment)

2026-05-21 Thursday 10:26 EDT initial; Friday 2026-05-22 ~10:38
EDT v0.4 prose absorption per Casey + Keeper textbook completion
phase

Vol 0 Chapter 1 — D_{IV}^5 Autogenic Proto-Geometry

Chapter motivation

Bubble Spacetime Theory begins with a single mathematical object: D_{IV}^5 , a 10-real-dimensional bounded Hermitian symmetric domain of complex dimension 5. This chapter introduces D_{IV}^5 to readers who may have encountered Lie groups, symmetric spaces, or complex geometry, and grounds the substrate object in standard mathematical literature.

D_{IV}^5 is not invented for BST. It is a classical object — studied since É. Cartan’s 1935 classification of irreducible Hermitian symmetric spaces (Cartan 1935; Helgason 1978 Theorem X.6.1; Wallach 1976; Faraut-Koranyi 1990/1994). BST identifies this specific geometry as substrate; the geometry exists independently of any physical interpretation.

By the end of this chapter, a reader should understand: 1. What D_{IV}^5 IS (Lie group quotient, Hermitian symmetric domain Type IV, complex dimension $n_C = 5$, real rank = 2) 2. Why this geometry (uniqueness criteria; preview of Vol 0 Ch 9 Strong-Uniqueness Theorem at 11 RIGOROUSLY CLOSED criteria, current ratified state per Calibration #19) 3. The naming convention (BST = the research framework, APG = “Autogenic Proto-Geometry” = D_{IV}^5 as substrate object) 4. How to compute on D_{IV}^5 (Bergman kernel, geodesics, group action, Wallach K-type decomposition)

Reader-grade pedagogy: this chapter is the substrate’s “what is this object” exposition. A graduate physicist with Lie group + bounded symmetric domain background should be able to read it linearly; a 5th-grader can follow the intuitive sum-

maries (BST identifies a specific 10-dimensional geometric object as the substrate underlying all physics; the geometry is the same one mathematicians studied since 1935; everything else in BST flows from this single starting point).

Diagram preview (full figures in v1.0): Section 1.1 will include (a) Cartan classification chart showing 4 types + 2 exceptional + D_{IV}^5 position; (b) Bergman kernel reproducing-property diagram for $f(z) \in H^2(D_{IV}^5)$; (c) $SO_0(5,2)$ action on D_{IV}^5 with isotropy subgroup decomposition $SO(5) \times SO(2)$; (d) compact dual quadric Q^5 as 5-quadric in CP^6 via Borel-Weil-Bott. These diagrams are pedagogical scaffolds; the underlying mathematics is in Sections 1.1-1.7.

Section 1.1 — The substrate object

$$D_{IV}^5 = SO_0(5,2) / [SO(5) \times SO(2)]$$

Read this expression as: the connected component of the special orthogonal group $SO(5,2)$ — orthogonal transformations preserving a (5+2)-dimensional indefinite inner product of signature (5,2) — modulo its isotropy subgroup $SO(5) \times SO(2)$.

Structural facts: - Real dimension: 10 ($= \dim SO_0(5,2) - \dim[SO(5) \times SO(2)] = 21 - 11$) - Complex dimension n_C : 5 (D_{IV}^5 is a complex manifold; n_C is BST primary integer) - Rank: 2 (the rank of the symmetric space; BST primary integer) - Type: IV in É. Cartan's classification (Type IV-n means orthogonal-signature domain of complex dimension n) - Classical name: Lie ball, generalized upper half-plane, type IV bounded symmetric domain

Bergman geometry: - D_{IV}^5 is a bounded symmetric domain — admits a canonical Bergman kernel + Bergman metric - Bergman kernel $K(z,w)$ reproduces holomorphic functions: $f(z) = \int K(z,w) f(w) dV(w)$ - Bergman metric is invariant under $SO_0(5,2)$ action — substrate-natural geometry - Bergman exponent $(g+\text{rank})/\text{rank} = 9/2$ (per BST primary identity)

Section 1.2 — Why this geometry

The full uniqueness argument is Vol 0 Chapter 9 (Strong-Uniqueness Theorem). This section previews the structural reasons:

1.2.1 Cartan classification context

É. Cartan classified irreducible Hermitian symmetric domains in 1935 into four types: - Type I: $D_{I_p,q} = SU(p,q)/[S(U_p \times U_q)]$ (complex matrix domains) - Type II: $D_{II_n} = SO^*(2n)/U(n)$ (skew-symmetric matrix domains) - Type III: $D_{III_n} = Sp(n,\mathbb{R})/U(n)$ (symmetric matrix domains) - Type IV: $D_{IV_n} = SO_0(n,2)/[SO(n) \times SO(2)]$ (orthogonal-signature domains)

Plus two exceptional domains: $E_6 / [Spin(10) \times U(1)]$ (dimension 16) + $E_7 / [E_6 \times U(1)]$ (dimension 27).

D_{IV}^5 is Type IV with $n = 5$. The Type IV family is the orthogonal-signature domain family — geometrically distinguished by its connection to conformal geometry in $(n+1)$ -dimensional spacetime ($SO_0(n,2)$ is the conformal group of Minkowski $_{(n-1,1)}$).

1.2.2 Why $n = 5$ within Type IV (preview)

Per Lyra T2431 (Why $n_C = 5$): - $n = 5$ within Type IV places D_{IV}^5 as 4-dimensional Minkowski + 1 extra dimension worth of substrate structure - Bergman exponent $9/2 = (g+rank)/rank$ requires specific structural parameter - Stark small-primary subset $\{-3, -7, -11\}$ substrate-anchoring requires $n_C = 5$ - $N_{max} = N_c^3 \cdot n_C + rank = 27 \cdot 5 + 2 = 137$ fixes n_C exactly

1.2.3 Why Type IV vs Types I/II/III/Exceptional (preview)

Per Strong-Uniqueness Theorem Vol 0 Ch 9 multi-criterion convergence: - Type IV has $SO_0(n,2)$ conformal symmetry $\rightarrow$ natural for Lorentz-structure physics - Types I/II/III have unitary or symplectic structure $\rightarrow$ less natural for physical spacetime - Exceptional E_6 (16D) or E_7 (27D) — wrong dimension for physical spacetime - Multi-criterion convergence (per Ch 9) singles out Type IV at $n = 5$

Type IV at $n = 5$ is the unique answer to the substrate uniqueness question.

Section 1.2b — Cross-Cartan EXHAUSTIVE at $dim_C = 5$ (T2455 Friday Lyra)

A foundational structural fact (T2455 Lyra Friday 2026-05-22): at $dim_C = 5$, the Helgason 1978 Theorem X.6.1 classification of irreducible bounded Hermitian symmetric domains produces EXACTLY THREE candidates:

$\{D_{IV}^5, D_{I_{\{1,5\}}}, D_{I_{\{5,1\}}}\}$

No other Cartan type produces $dim_C = 5$: - D_{II_n} : $n(n-1)/2 = 5 \rightarrow n(n-1) = 10$, no integer solution - D_{III_n} : $n(n+1)/2 = 5 \rightarrow n(n+1) = 10$, no integer solution - E_{III} : $dim_C = 16 \neq 5$ - E_{VII} : $dim_C = 27 \neq 5$

The cross-Cartan comparison at $dim_C = 5$ is therefore EXHAUSTIVE, not partial. Among these three HSD candidates, only D_{IV}^5 produces:

- Lowest non-trivial K-type Casimir = 6 ($= T_{\{N_c\}} =$ BST primary; $D_{I_{\{1,5\}}} = D_{I_{\{5,1\}}} = 4$)
- α -analog = 137 ($= N_{max}$ BST primary cap matching experimental $\alpha^{-1} = 137.036$ at 0.026%; $D_I = 41 \neq 137$ per T2456 universal formula)
- Bergman $c_{FK} = 225/\pi^{(9/2)}$ (BST primary form per T2442 RIGOROUSLY CLOSED)

D_{IV}^5 is uniquely-forced as substrate at its natural dimension $dim_C = 5 = n_C$ (BST primary) per Helgason 1978 classification.

Section 1.2c — Substrate Self-Amenability via n_C Primality (T2463 Friday Lyra)

A methodological structural observation (T2463 Lyra Friday): $n_C = 5$ (BST primary forced by T2445 Strong-Uniqueness C3 Thursday) IS PRIME. At PRIME $\dim_C$ values, the Helgason 1978 classification produces minimum HSD count (typically 3 candidates: $D_{IV_p} + D_{I_{\{1,p\}}} + D_{I_{\{p,1\}}}$ for prime p). At COMPOSITE $\dim_C$ values, more $D_{I_{\{p,q\}}}$ factorizations + possibly $D_{II_n} + D_{III_n}$ contributions appear.

At $\dim_C = 5$ prime: 3 HSDs total (Section 1.2b EXHAUSTIVE enumeration). At $\dim_C = 6$ composite: 7+ HSDs ($D_{IV_6} + D_{I_{\{1,6\}}} + D_{I_{\{6,1\}}} + D_{I_{\{2,3\}}} + D_{I_{\{3,2\}}} + D_{II_4} + D_{III_3}$).

The substrate's $n_C = 5$ PRIMALITY enables EXHAUSTIVE Cross-Cartan enumeration tractable at the substrate's natural dimension. The substrate is methodologically self-amenable to its own uniqueness verification: its BST primary choice $n_C = 5$ creates the conditions where the strong-uniqueness argument can be made EXHAUSTIVE at $\dim_C = n_C$.

Cross-link to Casey-named Substrate Working Process Principle (Tuesday): the substrate's operational efficiency includes methodological self-amenableity for uniqueness verification.

Section 1.3 — D_{IV}^5 as 4D Minkowski substrate

A key structural observation per Casey's research arc:

D_{IV}^5 contains 4-dimensional Minkowski spacetime as conformal boundary structure.

The isotropy subgroup $SO(5) \times SO(2)$: - $SO(5)$: spacetime rotation generators (5 independent directions in BST primaries) - $SO(2)$: internal symmetry generator (the $U(1)$ -equivalent factor responsible for electric charge per Casey W-56)

The full group $SO_0(5,2)$: - 21-dimensional Lie group - Conformal group of 5-dimensional Euclidean space (or equivalently, 4-dimensional Minkowski conformal group is $SO_0(4,2)$ — close to $SO_0(5,2)$) - Contains Poincaré group as subgroup

This is why BST is a Bubble Spacetime Theory — D_{IV}^5 provides the “bubble” of spacetime + internal structure together. The substrate is geometrically a bubble of conformal-symmetric 10-dimensional structure that produces 4D Minkowski + internal symmetries as observable structure.

Section 1.4 — Naming convention (BST vs APG)

Per notes/BST_Autogenic_Proto_Geometry_Definition.md (T1427):

- BST (Bubble Spacetime Theory): the theory, the research program, the physical predictions, the experimental falsifiers. Use BST for outreach to physicists, papers about physical predictions, talks about the framework's empirical claims.
- APG (Autogenic Proto-Geometry): the geometric object D_{IV}^5 itself. Use APG when discussing the geometry as mathematical object, formal uniqueness theorems, papers for mathematicians, definitions.

Rule of thumb: WHAT it IS $\rightarrow$ APG. WHAT it DOES $\rightarrow$ BST.

Why “Autogenic”: D_{IV}^5 generates its own structure — the BST primary integers (rank, N_c , n_C , C_2 , g , N_{max}) emerge from the geometry itself, not from external postulates. “Autogenic” = self-generating. The geometry is genetically self-defining.

Why “Proto”: D_{IV}^5 is the substrate from which all other physical structures emerge. “Proto” = first, foundational. The geometry comes BEFORE physics in derivation order.

Why “Geometry”: it's a Hermitian symmetric domain with explicit metric, curvature, holomorphic structure. Standard differential geometry tools apply.

Section 1.5 — How to compute on D_{IV}^5

For physicists/mathematicians wanting to operationally work with D_{IV}^5 :

1.5.1 Realization as bounded domain

D_{IV}^5 has a standard realization as a bounded domain in $\mathbb{C}^5$: - Domain $D_{IV}_n = \{z = (z_1, \dots, z_n) \in \mathbb{C}^n : 1 - 2|z|^2 + |z \cdot z|^2 > 0, |z|^2 < 1\}$ - For $n_C = 5$: $D_{IV}^5 \subset \mathbb{C}^5$ bounded by specific polynomial inequality - Center: origin $z = 0$ - Boundary: Shilov boundary at $|z| = 1$ with additional structure - Bergman kernel: explicit formula (Faraut-Koranyi 1994)

1.5.2 Group action

$SO_0(5,2)$ acts on D_{IV}^5 by holomorphic isometries. The action is transitive (any point can be mapped to origin).

1.5.3 Bergman kernel explicit form

For D_{IV}^5 , the Bergman kernel $K(z, \bar{w})$ has a specific polynomial form derived from the Faraut-Koranyi factorization. Per Lyra T2403 (Wednesday cascade-unblock): - $c_{FK} \cdot \pi^{(9/2)} = (N_c \cdot n_C)^2 = 225$ EXACT - c_{FK} is the Faraut-Koranyi normalization constant - $\pi^{(9/2)}$ emerges as Bergman exponent $(g+rank)/rank \cdot \pi$

This explicit formula anchors substrate calculations.

1.5.4 Geodesics + curvature

D_{IV}^5 has constant negative holomorphic sectional curvature (per bounded symmetric domain structure). Geodesics are well-studied; explicit formulas in Faraut-Koranyi 1994.

Section 1.6 — Connection to other curriculum chapters

D_{IV}^5 as substrate object grounds every other Vol 0 chapter + every Vol 1-10 curriculum chapter:

- Vol 0 Ch 2 (Five Integers + $N_{\max}$): the integers emerge from D_{IV}^5 structure
- Vol 0 Ch 3 (Substrate Operating System): commitment cycle + Koons tick + Reed-Solomon coding all operate on D_{IV}^5
- Vol 0 Ch 4 (Isotropy Group Structure): $SO(5) \times SO(2) \times$ Möbius decomposition
- Vol 0 Ch 5 (Boundary Conditions): D_{IV}^5 bulk + Shilov boundary + 6+2 BC framework
- Vol 0 Ch 6 (Integer Web Principle): each BST primary integer holds a web on D_{IV}^5
- Vol 0 Ch 7 (The Operator Zoo): operators act on Bergman $H^2(D_{IV}^5)$ per Lyra SP-31-1
- Vol 0 Ch 8 (Conservation Laws): Noether on substrate symmetries of D_{IV}^5
- Vol 0 Ch 9 (Strong-Uniqueness Theorem): why D_{IV}^5 specifically vs alternatives
- Vol 0 Ch 10 (Methodology Stack): audit-chain governance of D_{IV}^5 -derived claims

Every chapter reduces to “what does D_{IV}^5 say about X?” for some specific X.

Section 1.7 — Bergman H^2 (substrate Hilbert space)

Per Lyra SP-31-1 v0.1 (Thursday morning, paper-grade Cal #69 PASS):

Canonical substrate Hilbert space: Bergman $H^2(D_{IV}^5)$ = space of holomorphic functions on D_{IV}^5 square-integrable with respect to Bergman volume.

Three-layer hierarchy (Lyra T2428 + T2429 + T2430): - Bergman H^2 = integrated-state space (continuum physics) - $GF(128)^k$ Reed-Solomon code-space = per-Koons-tick discretization (T2429 corollary) - L^2 -section equivariant complement = representation-theoretic complement (T2430 corollary)

All three derived from one canonical anchor; none compete.

This is the Hilbert space all substrate-native operators (Vol 0 Ch 7) act on. The substrate Hilbert space anchors all subsequent quantum mechanics derivations in Vol 1.

Section 1.8 — BST ↔ standard physics dictionary entries

Standard math/physics term	BST term	Reference
Hermitian symmetric domain Type IV	D_IV ⁵ Autogenic Proto-Geometry (APG)	§1.1
Bergman kernel	Substrate kernel (defines substrate Hilbert space)	§1.7
Lie group SO ₀ (5,2)	Substrate symmetry group (conformal action on bubble spacetime)	§1.1
Hermitian symmetric domain	“the geometry”	substrate object
Bounded symmetric domain realization	Operational computation surface	§1.5
Cartan classification	Vol 0 Ch 9 uniqueness argument context	§1.2.1

Section 1.9 — Chapter status summary

Coverage at v0.1: - §1.1 D_IV⁵ as Lie group quotient with classical math context - §1.2 Why this geometry (Cartan + n_C + Type IV preview of Ch 9) - §1.3 4D Minkowski substrate framing (bubble spacetime origin) - §1.4 BST/APG naming convention - §1.5 Operational computation primitives - §1.6 Connection to other curriculum chapters - §1.7 Bergman H² substrate Hilbert space - §1.8 BST ↔ standard physics dictionary

Believability: classical math (Cartan classification, Bergman geometry, Faraut-Koranyi factorization) is recognizable to mathematicians and mathematical physicists. BST-specific framing (substrate, APG, bubble spacetime) introduced gradually. No advanced BST claims in Ch 1 — pure foundation.

Provability: classical math references (Cartan 1935, Bergman 1922, Faraut-Koranyi 1994, Wallach 1976) + T1427 BST_Autogenic_Proto_Geometry_Definition.md + Lyra SP-31-1 + T2403 c_FK · $\pi^{(9/2)} = 225$ EXACT. No new theorems introduced in Ch 1; this is the foundation chapter.

Path to v1.0: requires Lyra theoretical refinement of Bergman kernel explicit formula + Cal dual-axis grade-pass + integration with Vol 0 Ch 2-10 once all chapters at chapter-grade. Multi-week to multi-month evolution.

Per Casey’s standard

- Simple: D_IV⁵ is a specific Hermitian symmetric domain studied since Cartan 1935

- Works: BST identifies this geometry as substrate; all subsequent BST derivations work on D_{IV}^5
- Hard to break: would require finding D_{IV}^5 doesn't exist (mathematically impossible — Cartan classified it) OR finding the geometry doesn't admit the claimed structure (verified via Bergman + Faraut-Koranyi + Wallach references)

Status

Vol 0 Chapter 1 v0.1 chapter-grade content draft FILED Thursday 2026-05-21 10:26 EDT. Fourth Keeper-lane chapter-grade content (after Ch 8 + Ch 9 + Ch 10). FOUNDATIONAL CHAPTER introducing substrate object D_{IV}^5 . Classical math context + BST/APG naming + operational computation + connection to other curriculum chapters + Bergman H^2 substrate Hilbert space. Awaits Cal dual-axis grade-pass + Lyra theoretical refinement for v0.2.

— Keeper, 2026-05-21 Thursday 10:26 EDT (actual via date)